THE CHINESE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH1025/1028 Essential Mathematical Methods 2025-26 First Term

Solution to Homework 9

Complusory Questions

1. Suppose that $\mathbf{m}, \mathbf{n} \in \mathbb{R}^3$, where $|\mathbf{m}| = 2$, $|\mathbf{n}| = 1$ and the angle between $\mathbf{m}$ and $\mathbf{n}$ is $\frac{2\pi}{3}$.

   If $\mathbf{p} = 3\mathbf{m} + 4\mathbf{n}$ and $\mathbf{q} = 2\mathbf{m} - \mathbf{n}$, find

   (a) $\mathbf{m} \cdot \mathbf{n}$,

   (b) $|\mathbf{p}|$ and $|\mathbf{q}|$,

   (c) the area of the parallelogram spanned by $\mathbf{p}$ and $\mathbf{q}$.

   Ans:

   (a) $\mathbf{m} \cdot \mathbf{n} = |\mathbf{m}||\mathbf{n}| \cos(\frac{2\pi}{3}) = -1$

   (b) $|\mathbf{p}|^2 = \mathbf{p} \cdot \mathbf{p} = (3\mathbf{m} + 4\mathbf{n}) \cdot (3\mathbf{m} + 4\mathbf{n}) = 9|\mathbf{m}|^2 + 24\mathbf{m} \cdot \mathbf{n} + 16|\mathbf{n}|^2 = 28$. Therefore, $|\mathbf{p}| = 2\sqrt{7}$. Similarly, $|\mathbf{q}|^2 = \mathbf{q} \cdot \mathbf{q} = (2\mathbf{m} - \mathbf{n}) \cdot (2\mathbf{m} - \mathbf{n}) = 4|\mathbf{m}|^2 - 4\mathbf{m} \cdot \mathbf{n} + |\mathbf{n}|^2 = 21$. Therefore, $|\mathbf{q}| = \sqrt{21}$.

   (c) We have $\mathbf{p} \cdot \mathbf{q} = (3\mathbf{m} + 4\mathbf{n}) \cdot (2\mathbf{m} - \mathbf{n}) = 15$.

   Let the angle between $\mathbf{p}$ and $\mathbf{q}$ be θ. Then $\cos\theta = \frac{\mathbf{p} \cdot \mathbf{q}}{|\mathbf{p}||\mathbf{q}|} = \frac{15}{14\sqrt{3}}$. Therefore, $\sin\theta = \frac{11}{14}$.

   The area of the parallelogram spanned by $\mathbf{p}$ and $\mathbf{q}$ is $|\mathbf{p}||\mathbf{q}| \sin\theta = 11\sqrt{3}$.

2. Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be the position vectors of points A, B and C in $\mathbb{R}^3$, i.e. $\overrightarrow{OA} = \mathbf{a}$ and so on. Given that

$$\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} = \mathbf{0},$$

   show that A, B and C are collinear.

   Ans:

$$\begin{aligned}
\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} &= \mathbf{0} \\
\mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{c} &= \mathbf{0} \\
\mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{c} + \mathbf{a} \times \mathbf{a} &= \mathbf{0} \\
\mathbf{b} \times (\mathbf{c} - \mathbf{a}) - \mathbf{a} \times (\mathbf{c} - \mathbf{a}) &= \mathbf{0} \\
(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) &= \mathbf{0} \\
\overrightarrow{AB} \times \overrightarrow{AC} &= \mathbf{0}
\end{aligned}$$

   Therefore, A, B and C are collinear.

3. Let $\overrightarrow{OA} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\overrightarrow{OB} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\overrightarrow{OC} = 5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$.

   (a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$.

   (b) Find the volume of tetrahedron $OABC$.

   (Hint: Its volume equals to $\frac{1}{6}\times$volume of parallelotope spanned by $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$.)

   (c) By (a) and (b), find the distance from O to $\triangle ABC$.

   Ans:

   (a) Firstly, we have $\overrightarrow{AB} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\overrightarrow{AC} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Then,

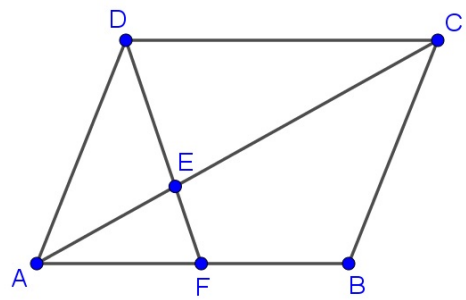
$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 4 & -1 & 2 \end{vmatrix} = -\mathbf{i} + 2\mathbf{k}.$$

(b) $(\overrightarrow{OA} \times \overrightarrow{OB}) \cdot \overrightarrow{OC} = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 5 & 1 & 3 \end{vmatrix} = 1.$

Therefore, the volume of tetrahedron $OABC = \frac{1}{6} \times |(\overrightarrow{OA} \times \overrightarrow{OB}) \cdot \overrightarrow{OC}| = \frac{1}{6}$.

(c) From (a), the area of $\triangle ABC = \frac{1}{2} \times |\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{\sqrt{5}}{2}$.
Let h be the distance from O to $\triangle ABC$.
Note that h is just the height of the tetrahedron $OABC$ with base $\triangle ABC$.
Then, $\frac{1}{3} \times \frac{\sqrt{5}}{2} \times h = \frac{1}{6}$ and so $h = \frac{1}{\sqrt{5}}$.

4.



In the above diagram, $ABCD$ is a parallelogram and F is a point on AB.
Suppose that DF and AC intersect at the point E such that $DE : EF = \lambda : 1$, where $\lambda > 0$.
Let $\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{AD} = \mathbf{q}$, $\overrightarrow{AE} = h\overrightarrow{AC}$ and $\overrightarrow{AF} = k\overrightarrow{AB}$, where $h, k > 0$.

(a) i. Express $\overrightarrow{AE}$ in terms of h, $\mathbf{p}$ and $\mathbf{q}$.
ii. Express $\overrightarrow{AE}$ in terms of λ, k $\mathbf{p}$ and $\mathbf{q}$.
Hence, show that $\lambda = \frac{1}{k}$.

(b) Given that $|\mathbf{p}| = 3$, $|\mathbf{q}| = 2$ and $\angle DAB = \frac{\pi}{3}$.
i. Find $\mathbf{p} \cdot \mathbf{q}$.
ii. Suppose that DF is perpendicular to AC.
(1) Express $\overrightarrow{DF}$ in terms of k, $\mathbf{p}$ and $\mathbf{q}$, and so find the value of k.
(2) Using (a), find the length of AE.

Ans:

(a) i. $\overrightarrow{AE} = h\overrightarrow{AC} = h(\mathbf{p} + \mathbf{q})$
ii. $\overrightarrow{AE} = \frac{1}{1+\lambda}(\lambda\overrightarrow{AF} + 1\overrightarrow{AD}) = \frac{1}{1+\lambda}(k\lambda\mathbf{p} + \mathbf{q})$.
Therefore, $h(\mathbf{p} + \mathbf{q}) = \frac{1}{1+\lambda}(k\lambda\mathbf{p} + \mathbf{q})$. Then, we have $h = \frac{k\lambda}{1+\lambda}$ and $h = \frac{1}{1+\lambda}$.
Hence, $k\lambda = 1$ and the result follows.

(b) i. $\mathbf{p} \cdot \mathbf{q} = |\mathbf{p}||\mathbf{q}| \cos \angle DAB = 3$
ii. (1) $\overrightarrow{DF} = \overrightarrow{AF} - \overrightarrow{AD} = k\mathbf{p} - \mathbf{q}$.
Since $\overrightarrow{DF} \perp \overrightarrow{AC}$, we have $\overrightarrow{DF} \cdot \overrightarrow{AC} = 0$. Then,

$$\begin{aligned} (k\mathbf{p} - \mathbf{q}) \cdot (\mathbf{p} + \mathbf{q}) &= 0 \\ k|\mathbf{p}|^2 + (k-1)\mathbf{p} \cdot \mathbf{q} - |\mathbf{q}|^2 &= 0 \\ 9k + 3(k-1) - 4 &= 0 \\ k &= \frac{7}{12} \end{aligned}$$

(2) For $k = \frac{7}{12}$, $\lambda = \frac{12}{7}$, $h = \frac{7}{19}$. Then, $|\overrightarrow{AE}|^2 = |\frac{7}{19}(\mathbf{p}+\mathbf{q})|^2 = (\frac{7}{19})^2(\mathbf{p}+\mathbf{q})\cdot(\mathbf{p}+\mathbf{q}) = \frac{49}{19}$.

Hence, $|\overrightarrow{AE}| = \frac{7\sqrt{19}}{19}$.

5. Let $L : x+1 = y+4 = \frac{z-2}{-2}$ be a straight line and let $\Pi : 2x+3y-z-5 = 0$ be a plane in $\mathbb{R}^3$.

   (a) Find the intersection point P of L and Π.

   (b) Let Π' be the plane in $\mathbb{R}^3$ such that Π' is normal to L and contains P. Find the equation of Π'.

   Ans:

   (a) Put $x = -1+t$, $y = -4+t$ and $z = 2-2t$ into the equation of Π, then

$$\begin{aligned} 2(-1+t)+3(-4+t)-(2-2t)-5 &= 0 \\ 7t &= 21 \\ t &= 3 \end{aligned}$$

   Therefore, the intersection point of L and Π is $(2,-1,-4)$.

   (b) Note that $\mathbf{n} = (1,1,-2)$ gives a normal vector of Π'. Therefore, the equation of Π' is $x+y-2z+D = 0$ for some $D \in \mathbb{R}$. Put $x = 2$, $y = -1$ and $z = -4$ into the equation,

$$\begin{aligned} (2)+(-1)-2(-4)+D &= 0 \\ D &= -9 \end{aligned}$$

   The equation of Π' is $x+y-2z-9 = 0$.

6. Suppose that $\Pi_1 : x+y+z = 1$ and $\Pi_2 : x-y+z = 2$ are two planes in $\mathbb{R}^3$.

   (a) Show that the intersection of Π_1 and Π_2 is a straight line and find a parametric equation of that line.

   (b) Find the equation(s) of the plane(s) containing all the points which are equidistant from Π_1 and Π_2.

   Ans:

   (a) We have

$$\begin{cases} x+y+z &= 1 \\ x-y+z &= 2 \end{cases}$$

$$\begin{cases} x+y+z &= 1 \\ -2y &= 1 \end{cases}$$

   Then, we have $y = -1/2$. If we put $y = -1/2$ into the first equation, we have $x+z = 3/2$. Let $z = t \in \mathbb{R}$, then $x = 3/2 - t$.
   Therefore, $(x,y,z) = (\frac{3}{2}-t, -\frac{1}{2}, t)$ is an intersection point of Π_1 and Π_2 for any $t \in \mathbb{R}$, i.e. Π_1 and Π_2 intersect at a straight line with a parametric equation $(x,y,z) = (\frac{3}{2}-t, -\frac{1}{2}, t)$, where $t \in \mathbb{R}$.

   (b) Suppose that $P = (x,y,z)$ is a point in $\mathbb{R}^3$ such that the distance between P and Π_1 and the distance between P and Π_2 are the same. Then, we have

$$\begin{aligned} \left|\frac{x+y+z-1}{\sqrt{1^2+1^2+1^2}}\right| &= \left|\frac{x-y+z-2}{\sqrt{1^2+(-1)^2+1^2}}\right| \\ |x+y+z-1| &= |x-y+z-2| \\ x+y+z-1 &= \pm(x-y+z-2) \end{aligned}$$

   The required planes are $2y = -1$ and $2x+2z = 3$.

Challenge Questions

7. Let $L_1 : \dfrac{x-5}{2} = \dfrac{y-1}{2} = \dfrac{z}{-1}$ and $L_2 : \dfrac{x-4}{2} = \dfrac{y+8}{5} = \dfrac{z-1}{2}$ be two straight lines in $\mathbb{R}^3$.

   (a) Show that L_1 and L_2 do not intersect.

   (b) Let L be the straight line perpendicular to L_1 and L_2, let A be the intersection point of L and L_1, and let B be the intersection point of L and L_2.

      (i) Find the coordinates of A and B.

      (ii) Find the equation of L.

   (c) Let Π be the plane containing the point A and perpendicular to L_1.

      (i) Find the equation of Π.

      (ii) Show that B lies on Π.

      (iii) Find the equation of the projection of L_2 on Π.

Ans:

(a) Rewrite the equations of L_1 and L_2 in parametric forms:

$$\begin{aligned} L_1 : &\quad x = 5 + 2s, y = 1 + 2s, z = -s \\ L_2 : &\quad x = 4 + 2t, y = -8 + 5t, z = 1 + 2t \end{aligned}$$

where $s, t \in \mathbb{R}$.
By substituting $s = -1 - 2t$ into $5 + 2s = 4 + 2t$, we have $t = -\frac{1}{6}$.
By substituting $s = -1 - 2t$ into $1 + 2s = -8 + 5t$, we have $t = \frac{7}{9}$.
Therefore, L_1 and L_2 do not intersect.

(b) $\vec{d_1} = (2, 2, -1)$ and $\vec{d_2} = (2, 5, 2)$ are direction vectors of L_1 and L_2 respectively.
$\overrightarrow{OA} = (5 + 2s_0, 1 + 2s_0, -s_0)$
$\overrightarrow{OB} = (4 + 2t_0, -8 + 5t_0, 1 + 2t_0)$
$\overrightarrow{BA} = (1 + 2s_0 - 2t_0, 9 + 2s_0 - 5t_0, -1 - s_0 - 2t_0)$

Since $\overrightarrow{BA} \perp \vec{d_1}$,
$(1 + 2s_0 - 2t_0, 9 + 2s_0 - 5t_0, -1 - s_0 - 2t_0) \cdot (2, 2, -1) = 0$ and $3s_0 - 4t_0 + 7 = 0$

Since $\overrightarrow{BA} \perp \vec{d_2}$,
$(1 + 2s_0 - 2t_0, 9 + 2s_0 - 5t_0, -1 - s_0 - 2t_0) \cdot (2, 5, 2) = 0$ and $4s_0 - 11t_0 + 15 = 0$

Therefore, $s_0 = -1$ and $t_0 = 1$.

(i) $A = (3, -1, 1)$ and $B = (6, -3, 3)$.

(ii) $\overrightarrow{BA} = (-3, 2, -2)$ and A lies on L.
Equation of $L : \dfrac{x-3}{3} = \dfrac{y+1}{-2} = \dfrac{z-1}{2}$.

(c) (i) Let $ax + by + cz + d = 0$ be the equation of Π.
Since Π is perpendicular to L_1, $(a, b, c) = (2, 2, -1)$.
Since A lies on Π, $2(3) + 2(-1) + (-1)(1) + d = 0$ and $d = -3$.
Hence, the equation of Π is $2x + 2y - z - 3 = 0$.

(ii) Since $2(6) + 2(-3) - 3 - 3 = 0$, B lies on Π.

(iii) Let L_2' be the projection of L_2 on Π, and $\vec{d_2'}$ be the direction vector of L_2'. Denote the normal vector of Π as $\vec{n}$. Then $\vec{n} = (2, 2, -1)$. Recall $\vec{d_2} = (2, 5, 2)$.

The projection of $\vec{d_2}$ on Π is $\vec{d_2} - \dfrac{\vec{n} \cdot \vec{d_2}}{\vec{n} \cdot \vec{n}} \vec{n} = (2, 5, 2) - \dfrac{4}{3}(2, 2, -1)$.

We can take $\vec{d_2'} = 3(2, 5, 2) - 4(2, 2, -1) = (-2, 7, 10)$.

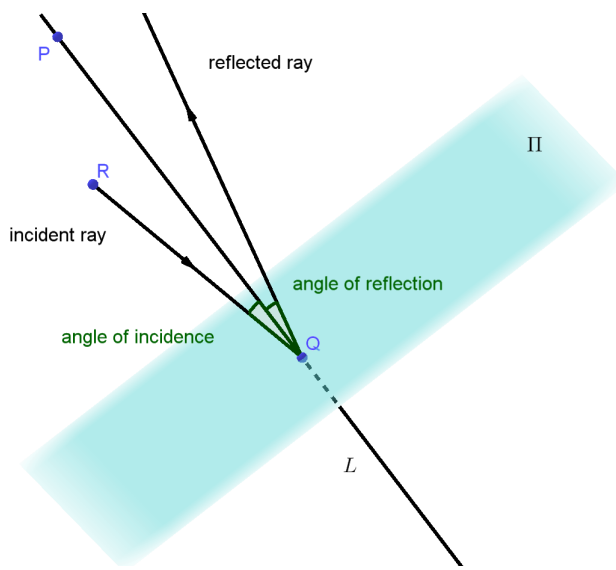
As $B = (6, -3, 3)$ lies on L_2', the equation of L_2' is $\dfrac{x-6}{-2} = \dfrac{y+3}{7} = \dfrac{z-3}{10}$.

8. Let $\Pi : 2x + y + 3z = 4$ be a plane and let $P = (8, 0, 10)$ be a point in $\mathbb{R}^3$.

(a) Let L be the straight line orthogonal to Π which passes through P and let Q be the intersection point of L and Π. Find a parametric equation of L and the coordinates of Q.

(b) Let $R = (7, 0, 6)$ be a point in $\mathbb{R}^3$. Find the projection of R on L.

(c) Suppose that $\overrightarrow{RQ}$ is a light ray that strikes the mirror Π. Find a parametric equation of the straight line that contains the reflected ray.
(Hint: The incident ray, the reflected ray and L are lying on the same plane. The angle of incidence equals the angle of reflection.)



Ans:

(a) $L : \mathbf{r}(t) = (8, 0, 10) + t(2, 1, 3)$. Since Q is a point on L, $Q = (8 + 2t_0, t_0, 10 + 3t_0)$ for some $t_0 \in \mathbb{R}$. Furthermore, Q is a point on Π, so

$$\begin{aligned} 2(8 + 2t_0) + t_0 + 3(10 + 3t_0) &= 4 \\ 14t_0 + 42 &= 0 \\ t_0 &= -3 \end{aligned}$$

Therefore, $Q = (2, -3, 1)$.

(b) Let T be the projection of R on L and let $\mathbf{n} = (2, 1, 3)$ be a normal of Π. Then,

$$\begin{aligned} \overrightarrow{QT} &= \text{proj}_{\mathbf{n}}(\overrightarrow{QR}) \\ &= \frac{\overrightarrow{QR} \cdot \mathbf{n}}{|\mathbf{n}|^2}\mathbf{n} \\ &= \frac{(5, 3, 5) \cdot (2, 1, 3)}{14}(2, 1, 3) \\ &= (4, 2, 6) \end{aligned}$$

Note that $\overrightarrow{OT} = \overrightarrow{OQ} + \overrightarrow{QT}$, and so $T = (6, -1, 7)$.

(c) Suppose that R' is the reflection of R along L lying on the plane containing the incident ray, the reflected ray and L. Note that

$$\overrightarrow{QR'} = \overrightarrow{QR} + \overrightarrow{RR'} = \overrightarrow{QR} + 2\overrightarrow{RT} = (5,3,5) + 2(-1,-1,1) = (3,1,7).$$

The straight line containing the reflected ray is

$$\mathbf{u}(t) = \overrightarrow{OQ} + t\overrightarrow{QR'} = (2,-3,1) + t(3,1,7).$$

Optional Questions

9. Let $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

(a) Express $\mathbf{a} = \mathbf{u} + \mathbf{v}$ such that $\mathbf{u}$ is parallel to $\mathbf{b}$ and $\mathbf{v}$ is orthogonal to $\mathbf{b}$.

(b) Find the area of parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

Ans:

(a) We have

$$\mathbf{u} = \text{proj}_{\mathbf{b}}(\mathbf{a}) = \left(\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2}\right)\mathbf{b} = \frac{8}{9}(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$$

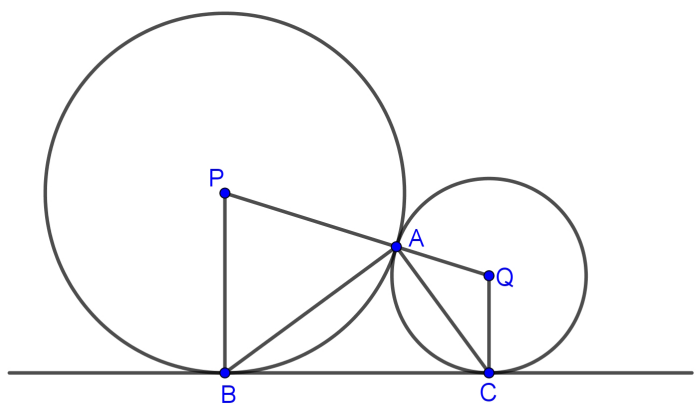
and

$$\mathbf{v} = \mathbf{a} - \mathbf{u} = \frac{1}{9}(28\mathbf{i} - 43\mathbf{j} + 29\mathbf{j}).$$

(b) The area of parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ is

$$|\mathbf{a}\times\mathbf{b}| = |\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -3 & 5 \\ 1 & 2 & 2 \end{vmatrix}| = |-16\mathbf{i} - 3\mathbf{j} + 11\mathbf{k}| = \sqrt{386}.$$

10. Two circles with centres at P and Q touch externally at the point A. BC is an exterior common tangent to the circles. Let $\overrightarrow{PA} = \mathbf{a}$, $\overrightarrow{PB} = \mathbf{b}$ and $QA : AP = k : 1$.



(a) Express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and k.

(b) Hence, show that AB is perpendicular to AC.

Ans:

(a) $\overrightarrow{AB} = \overrightarrow{PB} - \overrightarrow{PA} = \mathbf{b} - \mathbf{a}$.

Since $QA : AP = k : 1$, $\overrightarrow{AQ} = k\overrightarrow{PA} = k\mathbf{a}$. Furthermore, $QA = QC$ and $PA = PB$, so $QC : PB = k : 1$. Note that both QC and PB are perpendicular to the common tangent line, so $\overrightarrow{QC}$ is parallel to $\overrightarrow{PB}$ which gives $\overrightarrow{QC} = k\overrightarrow{PB} = k\mathbf{b}$. As a result, $\overrightarrow{AC} = \overrightarrow{AQ} + \overrightarrow{QC} = k(\mathbf{a}+\mathbf{b})$.

(b) Note that

$$\begin{aligned}\overrightarrow{AB}\cdot\overrightarrow{AC} &= (\mathbf{b}-\mathbf{a})\cdot k(\mathbf{a}+\mathbf{b})\\ &= k(|\mathbf{a}|^2-|\mathbf{b}|^2)\\ &= 0\end{aligned}$$

The last equality holds since both PA and PB are radii of the same circle. Therefore, AB is perpendicular to AC.

11. (a) Let $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$. Show that

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=(\mathbf{c}\cdot\mathbf{a})\mathbf{b}-(\mathbf{c}\cdot\mathbf{b})\mathbf{a}.$$

(b) Let $\mathbf{a},\mathbf{u}\in\mathbb{R}^3$. Suppose that $\mathbf{u}$ is a unit vector. Prove that

$$\mathbf{a}=(\mathbf{a}\cdot\mathbf{u})\mathbf{u}+(\mathbf{u}\times\mathbf{a})\times\mathbf{u}.$$

Ans:

(a) Let $\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$, $\mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}$ and $\mathbf{c}=c_1\mathbf{i}+c_2\mathbf{j}+c_3\mathbf{k}$. Then,

$$\mathbf{a}\times\mathbf{b}=(a_2b_3-a_3b_2)\mathbf{i}+(-a_1b_3+a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}.$$

Therefore,

$$\begin{aligned}(\mathbf{a}\times\mathbf{b})\times\mathbf{c} &= \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_2b_3-a_3b_2 & -a_1b_3+a_3b_1 & a_1b_2-a_2b_1\\ c_1 & c_2 & c_3\end{vmatrix}\\ &= [(-a_1b_3+a_3b_1)c_3-(a_1b_2-a_2b_1)c_2]\mathbf{i}+[-(a_2b_3-a_3b_2)c_3+(a_1b_2-a_2b_1)c_1]\mathbf{j}+\\ &\qquad [(a_2b_3-a_3b_2)c_2-(-a_1b_3+a_3b_1)c_1]\mathbf{k}\\ &= [(a_2c_2+a_3c_3)b_1\mathbf{i}+(a_1c_1+a_3c_3)b_2\mathbf{j}+(a_1c_1+a_2c_2)b_3\mathbf{k}]-\\ &\qquad [(b_2c_2+b_3c_3)a_1\mathbf{i}+(b_1c_1+b_3c_3)a_2\mathbf{j}+(b_1c_1+b_2c_2)a_3\mathbf{k}]\\ &= [(a_1c_1+a_2c_2+a_3c_3)b_1\mathbf{i}+(a_1c_1+a_2c_2+a_3c_3)b_2\mathbf{j}+(a_1c_1+a_2c_2+a_3c_3)b_3\mathbf{k}]-\\ &\qquad [(b_1c_1+b_2c_2+b_3c_3)a_1\mathbf{i}+(b_1c_1+b_2c_2+b_3b_3)a_2\mathbf{j}+(b_1c_1+b_2c_2+b_3c_3)a_3\mathbf{k}]\\ &= (\mathbf{c}\cdot\mathbf{a})\mathbf{b}-(\mathbf{c}\cdot\mathbf{b})\mathbf{a}\end{aligned}$$

(b) Use the result in (a) and replace $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ by $\mathbf{u}$, $\mathbf{a}$ and $\mathbf{u}$ respectively. Then,

$$\begin{aligned}(\mathbf{u}\times\mathbf{a})\times\mathbf{u} &= (\mathbf{u}\cdot\mathbf{u})\mathbf{a}-(\mathbf{u}\cdot\mathbf{a})\mathbf{a}\\ &= \mathbf{a}-(\mathbf{u}\cdot\mathbf{a})\mathbf{a} \qquad (\because \mathbf{u}\cdot\mathbf{u}=1)\\ \mathbf{a} &= (\mathbf{a}\cdot\mathbf{u})\mathbf{u}+(\mathbf{u}\times\mathbf{a})\times\mathbf{u}\end{aligned}$$

12. Find the angle between planes $\Pi_1: 3x-4y+7z+13=0$ and $\Pi_2: 8x+3y-z+2=0$.

Ans:

Note that the angle between Π_1 and Π_2 is exactly the angle between the normals $\vec{n}_1=3\mathbf{i}-4\mathbf{j}+7\mathbf{k}$ and $\vec{n}_2=8\mathbf{i}+3\mathbf{j}-\mathbf{k}$ of them.

Therefore, the angle between Π_1 and $\Pi_2=\cos^{-1}\left(\dfrac{\vec{n}_1\cdot\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}\right)=\cos^{-1}\left(\dfrac{5}{74}\right)$.

13. Let $P=(-2,1,4)$ be a point and $L: x-4=\dfrac{y-3}{3}=z-3$ be a straight line in $\mathbb{R}^3$.

Suppose that Π_1 is the plane in $\mathbb{R}^3$ passing through P and normal to L, and that Π_2 is the plane containing L and P.

(a) Find the equations of Π_1 and Π_2.

(b) Find the equation of the straight line at the intersection of Π_1 and Π_2.

Ans:

(a) $\mathbf{n} = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$ a direction vector of L, which gives a normal vector to Π_1.
Let the equation of Π_1 be $x + 3y + z + D = 0$. Since $P = (-2, 1, 4)$ lies on Π_1,

$$\begin{aligned} (-2) + 3(1) + (4) + D &= 0 \\ D &= -5 \end{aligned}$$

Therefore, the equation of Π_1 is $x + 3y + z - 5 = 0$.
Note that $Q = (4, 3, 3)$ be a point on L. Then,

$$\mathbf{n}' = \mathbf{n} \times \overrightarrow{QP} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 1 \\ -6 & -2 & 1 \end{vmatrix} = 5\mathbf{i} - 7\mathbf{j} + 16\mathbf{k}$$

gives a normal vector of Π_2. Let the equation of Π_2 be $5x - 7y + 16z + D' = 0$. Since $Q = (4, 3, 3)$ lies on Π_2,

$$\begin{aligned} 5(4) - 7(3) + 16(3) + D' &= 0 \\ D' &= -47 \end{aligned}$$

Therefore, the equation of Π_2 is $5x - 7y + 16z + 47 = 0$.

(b) Let Q' be the projection of Q on Π_1.
Then, the intersection of Π_1 and Π_2 is the line passing through P and Q'.
Since Q' lies on L, $Q' = (4, 3, 3) + t(1, 3, 1) = (4 + t, 3 + 3t, 3 + t)$ for some $t \in \mathbb{R}$.
Furthermore, Q' lies on Π_1, so

$$\begin{aligned} (4 + t) + 3(3 + 3t) + (3 + t) - 5 &= 0 \\ t &= -1 \end{aligned}$$

Therefore, $Q' = (3, 0, 2)$. The required equation is $\mathbf{r}(t) = \overrightarrow{OP} + t\overrightarrow{PQ'} = (-2, 1, 4) + t(5, -1, -2)$.